

HealthConnect Clinic

Week 6 — Advanced Analytics & Decision Support

Validating and Deepening the Week 5 No-Show Drivers

Track: Data Analytics | Week 6

Project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Week 5 → Week 6 Transition

- Week 5 output: full EDA across six dimensions, 5 KPIs, an initial 6-panel dashboard, and 6 business insights — all descriptive/associational.
- Most important Week 5 result: three drivers stood out individually — booking lead time (24.8% → 60.5% no-show rate), distance to clinic (46.5% → 68.1%), and prior no-show history (43.5% → 68.8%).
- Main limitation identified: findings were purely descriptive (bucketed averages), with no statistical testing and no combined view across drivers.
- Relevant track: Data Science (baseline no-show prediction model, built on features shared at the end of Week 5).
- Week 6 focus: statistically validate the Week 5 drivers individually and jointly; build a combined risk-segmentation view; test whether the reminder effect changes for high-risk patients; refine the high-risk segment shared with Data Science.

Part 1 — Statistical Validation of Week 5 Drivers

Week 5 reported no-show rate differences across buckets of each variable. Chi-square tests were used to confirm whether these differences are statistically significant, and a logistic regression was used to test whether each driver still matters once the others are controlled for.

Chi-square tests (individual drivers vs. no-show)

Variable			Chi-square	df	p-value
Booking lead time			320.4	4	4.4e-68
Distance to clinic			22.4	4	1.7e-04
Previous no-shows			80.6	3	2.3e-17
Reminder sent			6.3	1	0.012
Appointment day			4.5	6	0.612

Table 1. Chi-square tests of independence for each Week 5 driver against appointment outcome.

All Week 5 drivers except appointment day are statistically significant — the patterns reported in Week 5 are real, not sampling noise. Appointment day is confirmed as not meaningfully related to attendance, consistent with the Week 5 conclusion.

Joint logistic regression

Variable	Coefficient	Odds ratio	p-value
Booking lead days	0.034	1.035	<0.001
Distance to clinic (km)	0.022	1.022	<0.001
Previous no-shows	0.378	1.460	<0.001
Reminder sent	-0.189	0.828	0.005
Waiting time (min)	-0.001	0.999	0.839

Table 2. Logistic regression of no-show status on all five candidate drivers simultaneously.

Booking lead time, distance, prior no-shows, and reminder status remain significant even after controlling for one another — each is an independent contributor, not a proxy for the others. Waiting time remains non-significant ($p = 0.839$), confirming it is not a driver even under this stricter joint test.

Business relevance: because the three main risk factors are independent contributors rather than redundant signals, a combined risk score should outperform any single-variable flag used in Week 5.

Part 2 — Compound Risk Segmentation

Week 5 flagged high-risk patients using a single rule ($\text{previous_no_shows} \geq 2 \rightarrow 10.6\%$ of appointments, 61.0% no-show rate, capturing 13.4% of all no-shows). Since Part 1 shows the three main drivers are independent, combining them identifies a segment that is both higher-risk and captures far more of the no-show problem.

Risk score definition (0–3): +1 point each for booking lead time > 14 days, distance > 10 km, and previous no-shows ≥ 1 .

No-Show Rate by Compound Risk Score

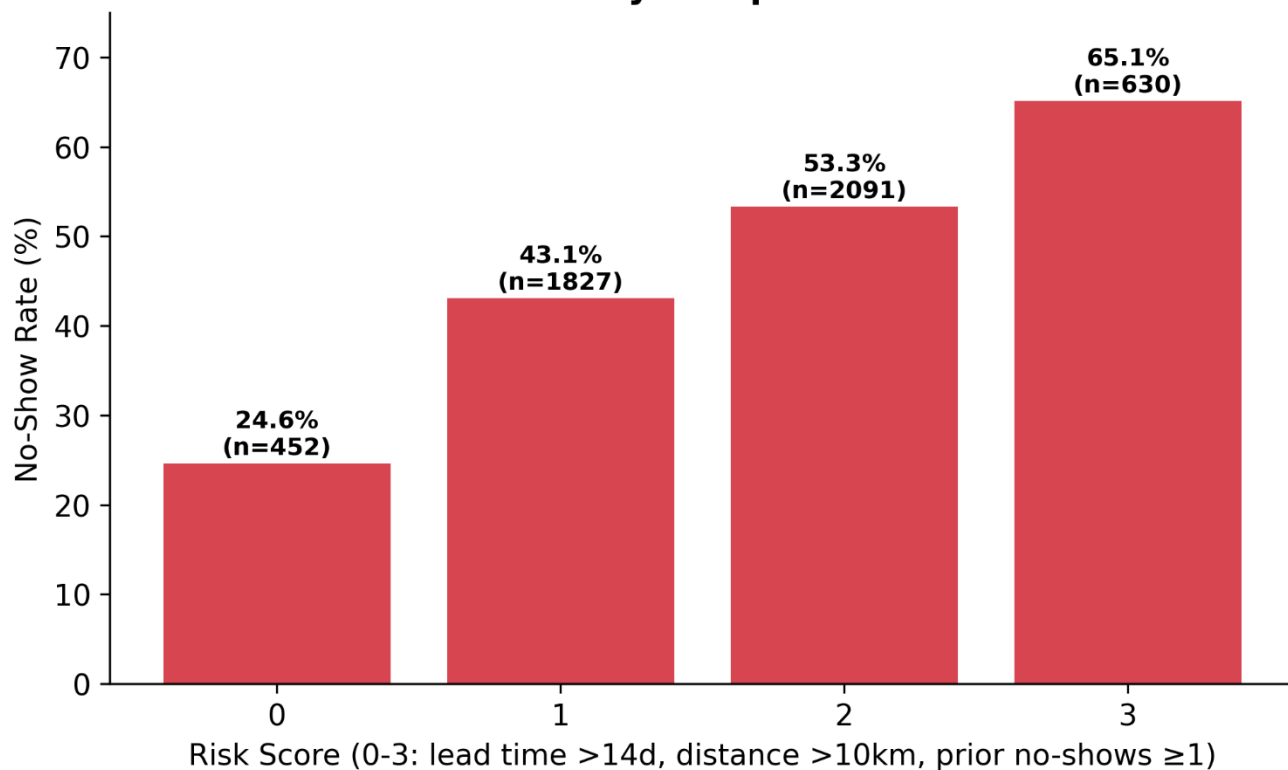


Figure 1. No-show rate by compound risk score (0–3).

Risk score	n	Share of appts	No-show rate	Share of all no-shows
0	452	9.0%	24.6%	4.6%
1	1827	36.5%	43.1%	32.5%
2	2091	41.8%	53.3%	46.0%
3	630	12.6%	65.1%	16.9%

Table 3. No-show rate and coverage at each risk score level.

No-show rate rises cleanly and monotonically with the risk score — from 24.6% (score 0) to 65.1% (score 3). This is a much stronger and more usable signal than any single driver on its own.

Comparison with the Week 5 definition

Figure 2. Coverage of the Week 5 single-variable rule vs. the Week 6 compound risk score.

The Week 6 compound definition (risk score ≥ 2) flags 54.4% of appointments with a 59.0% no-show rate, capturing 62.9% of all no-shows — versus 13.4% captured by the Week 5 rule. Because it flags a

larger share of the book, this should be used as a tiered intervention (heavier follow-up for score 3, lighter for score 2) rather than a single flag.

This refined segment definition is the primary update shared back with the Data Science track.

Part 3 — Does the Reminder Effect Hold for High-Risk Patients?

Week 5 found reminders reduce no-shows by about 4.0 percentage points overall. This section tests whether that effect changes for high-risk patients.

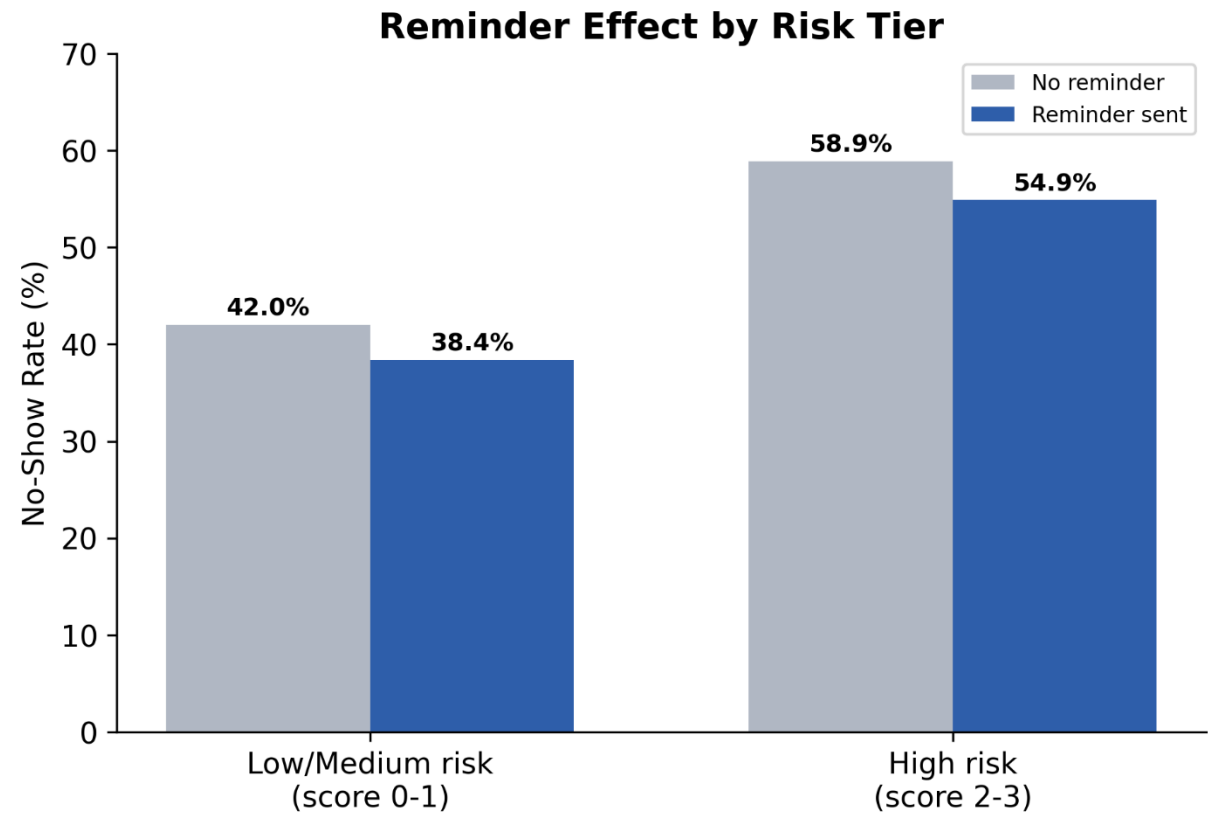


Figure 3. No-show rate with and without a reminder, split by risk tier.

Segment	No reminder	Reminder sent	Difference	p-value
Low/medium risk (score 0-1)	42.0%	38.4%	3.6 pp	0.132
High risk (score 2-3)	58.9%	54.9%	4.0 pp	0.067

Table 4. Reminder effect within each risk tier.

The reminder effect is similar in size for both groups, and neither difference reaches conventional significance at this segment size (high-risk group is borderline, $p = 0.067$). There is no evidence that reminders work meaningfully better or worse for high-risk patients — reminders should remain a universal baseline intervention, with the risk score used to decide who gets additional (non-reminder)

support such as a phone call, not who receives a reminder at all. This finding is flagged as suggestive rather than conclusive, and worth a properly powered test in Week 7.

Part 4 — Cross-Track Integration

Track collaborated with: Data Science.

Project dependency: Data Analytics' feature validation and risk segmentation inform Data Science's Week 6 feature refinement and error analysis; Data Science's model outputs (feature importance, error analysis) inform whether the Data Analytics risk segment needs further revision.

Information received from Data Science (Week 5 baseline): confirmation that booking_lead_days, distance_to_clinic_km, and previous_no_shows were used as baseline model features, consistent with the Week 5 hand-off.

Information provided to Data Science (this report):

- Statistical validation confirming all three features are significant individually and jointly — supports keeping them in the model rather than dropping any as redundant.
- Confirmation that waiting_time_minutes is not predictive even after controlling for other variables — supports excluding it as a model feature.
- The refined compound risk score, capturing 62.9% of no-shows in 54.4% of the book vs. 13.4%/10.6% for the Week 5 definition — usable as a benchmark for the trained model and as a sanity check for scoring new patients.
- The finding that reminder effectiveness does not vary meaningfully by risk tier — relevant if Data Science or GenAI explore reminder-related features or interventions.

What changed as a result: the high-risk segment definition shared with Data Science at the end of Week 5 is superseded by the Week 6 compound risk score, which is statistically grounded and captures roughly five times more of the no-show problem.

Evidence: this report, the accompanying Jupyter notebook (statistical tests, regression output, risk-segment tables and charts) — to be supplemented with the Data Science track's Week 6 feature-importance/error-analysis output once received.

Part 5 — Updated Findings, Recommendations & Limitations

Validated / refined findings

- Booking lead time, distance to clinic, and prior no-show history are statistically confirmed — individually and jointly — as the primary attendance drivers, not just descriptive patterns.
- Waiting time is confirmed as not predictive, even controlling for other factors.
- A compound risk score (0–3) produces a clean, monotonic gradient (24.6% → 65.1%) and substantially outperforms any single-variable rule used in Week 5.
- Reminder effectiveness is consistent across risk tiers — reminders should remain universal, with the risk score used to target additional interventions.

Updated recommendations

- Deploy the risk score (0–3) as the primary triage tool for administrative follow-up: light touch for score 1, proactive outreach for score 2, priority phone-call/rescheduling support for score 3.
- Keep SMS reminders universal (per Week 5) regardless of risk tier.
- Feed the risk score to the Data Science track as a benchmark for the predictive model, and revisit it once model-based feature importance is available.
- Continue treating distance-related recommendations as requiring approved clinic procedures (e.g. transport support), as noted in Week 5.

Limitations

- Dataset remains fictional/synthetic; all findings require validation against real operational data.
- Statistical significance confirms association, not causation — this remains an observational analysis.
- Risk score thresholds (14 days, 10 km, ≥ 1 prior no-show) were chosen by testing a small grid of options for a reasonable rate/coverage trade-off, not formally optimised — the Data Science model should be treated as the more rigorous version of this segmentation once available.
- The reminder-by-risk-tier test is borderline significant for the high-risk group ($p = 0.067$); a larger sample or A/B-style test is needed before treating this as conclusive.

Week 7 analytical testing requirements

- Compare the rule-based risk score against the Data Science model's predicted probabilities once available — quantify agreement/disagreement.
- Design a proper reminder-timing/channel test, stratified by risk tier, now that Part 3 suggests tier may not moderate the effect.
- Re-run the significance tests on any updated/expanded dataset before Week 7 validation.